Aaradhya Dev Tamrakar

AI Systems Engineering · Workflow Orchestration · Developer Tooling

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Final-year engineering student building AI workflow tooling, distributed agent coordination harnesses, and developer infrastructure. Developer of Fleet-Orchestrator (multi-step workflow coordination with provider adapters and task state inspection), Agent-Customization-Sync (cross-environment configuration and drift auditing), and custom Model Context Protocol (MCP) integrations.

EDUCATION

Bachelor of Engineering (B.E.) in Electronics, Communication and Information Engineering | December 2022 – Expected January 2027

Kathmandu Engineering College, Institute of Engineering (IOE), Tribhuvan University • Cumulative Average: ~75% across completed semesters

FELLOWSHIPS & APPLIED ML EXPERIENCE

Fuse AI Fellow — Fusemachines AI Fellowship 2026 | 2026 – Present (Active)

- Built applied agentic and ML coursework pipelines including a 5-stage Text-to-SQL system (Planner → Generator → Validator → Executor → Summarizer) tested against structured evaluation suites.
- Implemented MLOps coursework for experiment tracking, containerized microservices (**FastAPI**, **Docker**, **PostgreSQL**), and scheduled data-drift evaluation.

NSSR DataCamp Fellow (Cohort 2) — Nepalese Society of Student Researchers | April 2026 – Present

- Selected in competitive cohort for sponsored DataCamp access; developed applied GenAI workflows automating data cleaning, schema normalization, and exploratory analysis.

RESEARCH & TECHNICAL PROJECTS

BiasAperture — Computer Vision Demographic Fairness & Bias Audit | 2026

Python · AIF360 · Fairlearn · Statistical Testing · Jinja2 · OpenCV

Fleet Orchestrator — AI Workflow Coordination & Tooling | 2025 – Present

Python · FastAPI · Provider Adapters · Model Context Protocol (MCP) · SQLite

- Built coordination tooling for multi-step AI workflows, with task state, worker adapters, scheduling, and live status inspection.
- Standardized provider adapters (local models, CLI harnesses, API endpoints) with reusable interfaces to eliminate repeated orchestration scripts.
- Integrated local MCP tool servers and automated execution logging to ensure reproducible batch processing and reliable error recovery.

Agent-Customization-Sync — Cross-IDE Configuration Tooling | 2025 – Present

Python · SHA-256 · CLI · Configuration Governance

- Built configuration snapshots, guarded restore, rules transpilation, and SHA-256 drift audits across development environments (Antigravity, Cursor, Claude Code).
- Audited instruction overhead and linked shared skills to reduce configuration duplication and maintain environment consistency across machines.

Super-NLM — Multi-Account NotebookLM Aggregator & MCP Tool | 2025 – Present

Python · Model Context Protocol (MCP) · Google Drive API · FastAPI

- Developed an MCP tool server enabling AI agents to query notebooks, synchronize Google Drive research folders, and download Studio artifacts.
- Implemented queue scheduling and quota-aware backoff handling to support automated document synthesis across study materials.

Utility MCP Suite — LocalSend & Developer Bridges | 2025 – Present

Python · Model Context Protocol (MCP) · FastAPI · Networking

- Developed custom MCP servers including a LocalSend bridge enabling AI agents to discover LAN peers and transfer build artifacts securely.
- Built headless IPC connectors enabling agent-assisted operations with local design and engineering applications.

LEADERSHIP & SERVICE

Vice Chair (2026 – Present); Vice Secretary (2025 – 2026) — IEEE KEC KTM Student Branch | 2025 – Present

- Coordinate branch planning, technical workshops, and cross-committee communication for Kathmandu Engineering College IEEE student body.
- Facilitated technical awareness events and student engagement initiatives across engineering departments.

Makerspace Ambassador — KEC Maker's Space, Kathmandu Engineering College | June 2026 – Present

- Maintain website content, interface fixes, and documentation for student prototyping equipment.
- Support students with 3D printing, embedded electronics prototyping, and lab equipment access.

TECHNICAL SKILLS

AI Systems, Backend & Tooling: FastAPI, Docker, Model Context Protocol (MCP), SQL (PostgreSQL, SQLite), asyncio, REST APIs, MLflow

Engineering Practice & Tooling: Git, GitHub, Linux, PowerShell, Pytest, Reproducible Configurations, WordCraft, LaTeX